

WearUS: A Unified Framework for Wearable A-Mode Ultrasound Pretraining and Downstream Transfer

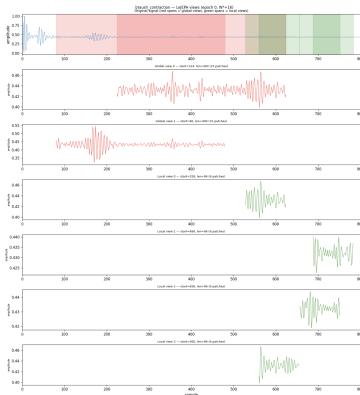
Fourteenth Weekly Report

Gabriele Merlino

Integrated Systems Laboratory (IIS)
Eidgenössische Technische Hochschule Zürich

June 16, 2026

LeJEPa Loss Formulation



Total LeJEPa Loss

The overall objective function is formulated as the sum of the similarity loss and the signature regularization term:

$$\mathcal{L}_{LeJEP A} = \mathcal{L}_{sim} + \lambda \mathcal{L}_{SigREG}$$

$$\lambda = 0.002$$

SIGReg 2D Projection (Epoch 399)

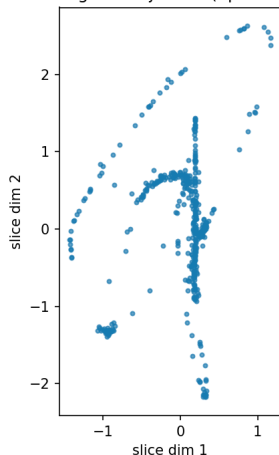


Figure: SigReg Projections

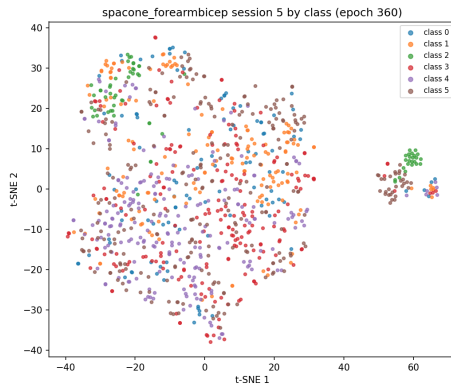


Figure: t-SNE: Downstream Task Embeddings

$$\lambda = 0.002$$

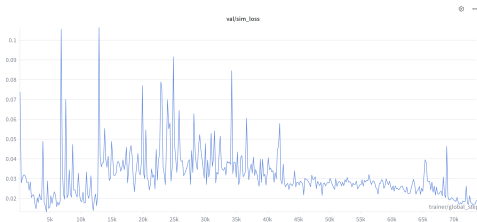


Figure: sim loss

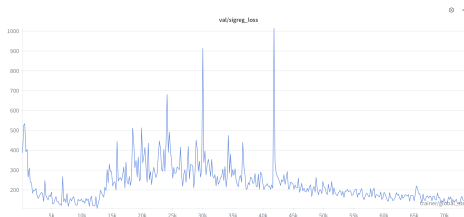


Figure: sigREG term

$$\lambda = 0.02$$

SIGReg 2D Projection (Epoch 399)

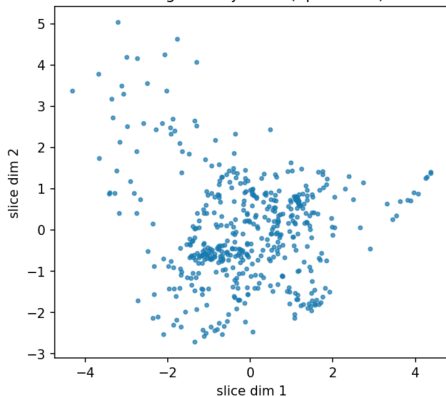


Figure: sigREG

spacone_forearmbicep session 5 by class (epoch 360)

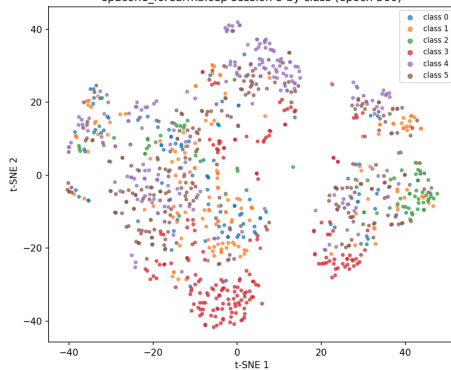


Figure: tsne

$$\lambda = 0.02$$



Figure: sim loss

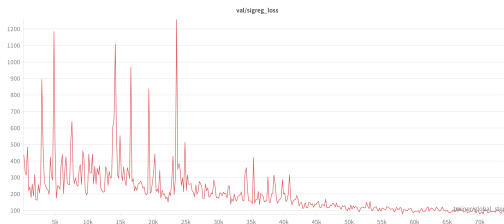


Figure: sigREG term

Results

Name	λ	Intra-session Acc.	Inter-session Acc.	# Parameters
LeJEPA finetuned	0.002	28%	22% $\pm$ 7%	50M
LeJEPA LinearProbing	0.002	27%	21.5% $\pm$ 5%	50M
LeJEPA finetuned	0.02	35%	25% $\pm$ 5%	50M
LeJEPA LinearProbing	0.02	35%	23% $\pm$ 4%	50M
From Scratch	—	98%	17% $\pm$ 4%	50M
MAE finetuned	—	100%	70.6% $\pm$ 7%	50M
Giusy CNN (my DataLoaders)	—	100%	65% $\pm$ 12%	150K

Note: All the results are relative to the usual wulpus dataset collected by Giusy.

Some ULTRAPRO results

Name	Intra-session Acc.	Inter-session Acc.	# Parameters
Best MAE	81.3%	22% $\pm$ 5%	50M
Giusy CNN (my DataLoaders)	83.5%	25% $\pm$ 7%	150K
Paper Authors	76.54%	—	500K